

TEAM MANGAUSSIAN

The Journey Navigation Flag: Turning Diabetes Encounters into Action

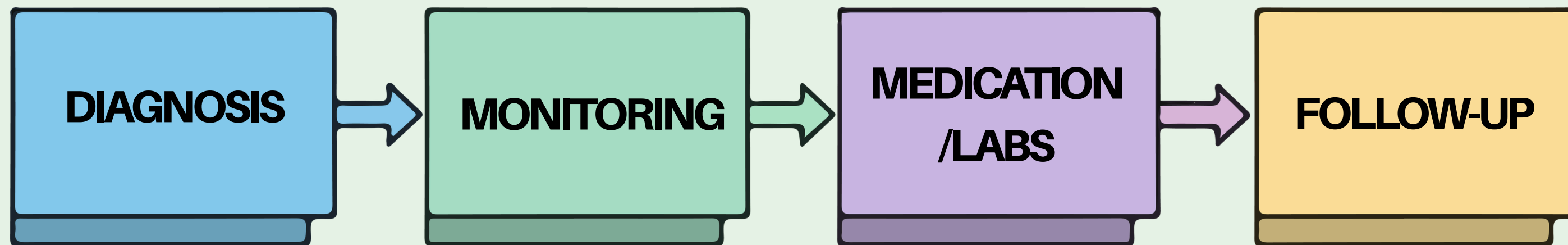
Many journeys are delayed or disconnected for Type 2 diabetes.
How can SVH utilize patients journeys to prioritize outreach early enough?

Presented by : **Zihang Peng, Seyoung Yoo**

Why Diabetes? Why Longitudinal?

- Health care is fundamentally longitudinal, by viewing data as snapshots we miss key information
- Diabetes is significant, it is the 2nd most diagnosed chronic disease
- Diabetes is where follow-up matters

CHRONIC CARE LOOP:



THE PROBLEM

THE FINDINGS

PREDICTIVE MODEL

CONCLUSION

What counts as a care gap?

Follow-up within 4 months = **timely, good baseline**

No follow-up within 4 months = **delayed gap / disconnection**

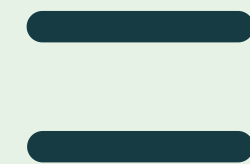
WHY 4 MONTHS?



A 2025 STUDY OF 89,674 TYPE 2 DIABETES PATIENTS FOUND THAT PATIENTS WITH >3 GP VISITS/YEAR (4 MONTHS GAP BETWEEN VISIT) WERE MORE LIKELY TO ACHIEVE BETTER DIABETES CONTROL

BOTTOM LINE:

MORE VISITS



BETTER OUTCOMES

GARCÍA-VILLARINO ET AL., PRIMARY CARE DIABETES, 2025

THE PROBLEM

THE FINDINGS

PREDICTIVE MODEL

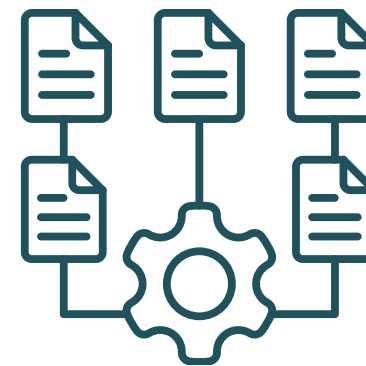
CONCLUSION

Methodology



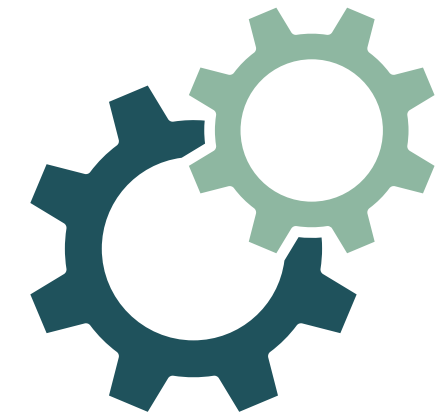
Raw information

- Encounters w/ SVH
- Patients
- Diagnosis
- Departments involved
- Providers involved
- Social determinants
- Census



Patient-diagnosis journey table

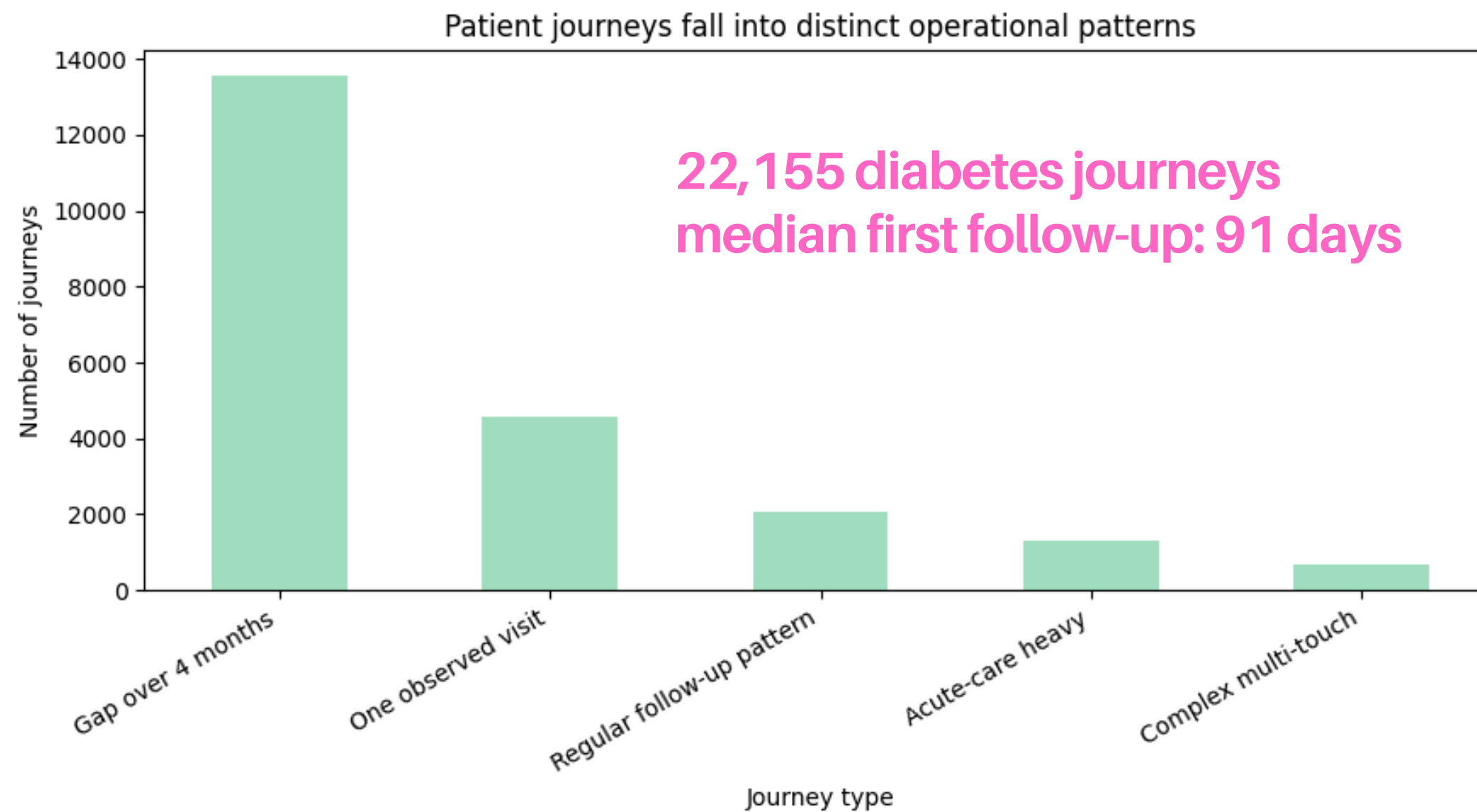
1 row = 1 patient's Type 2 diabetes journey



4 Output Variables:

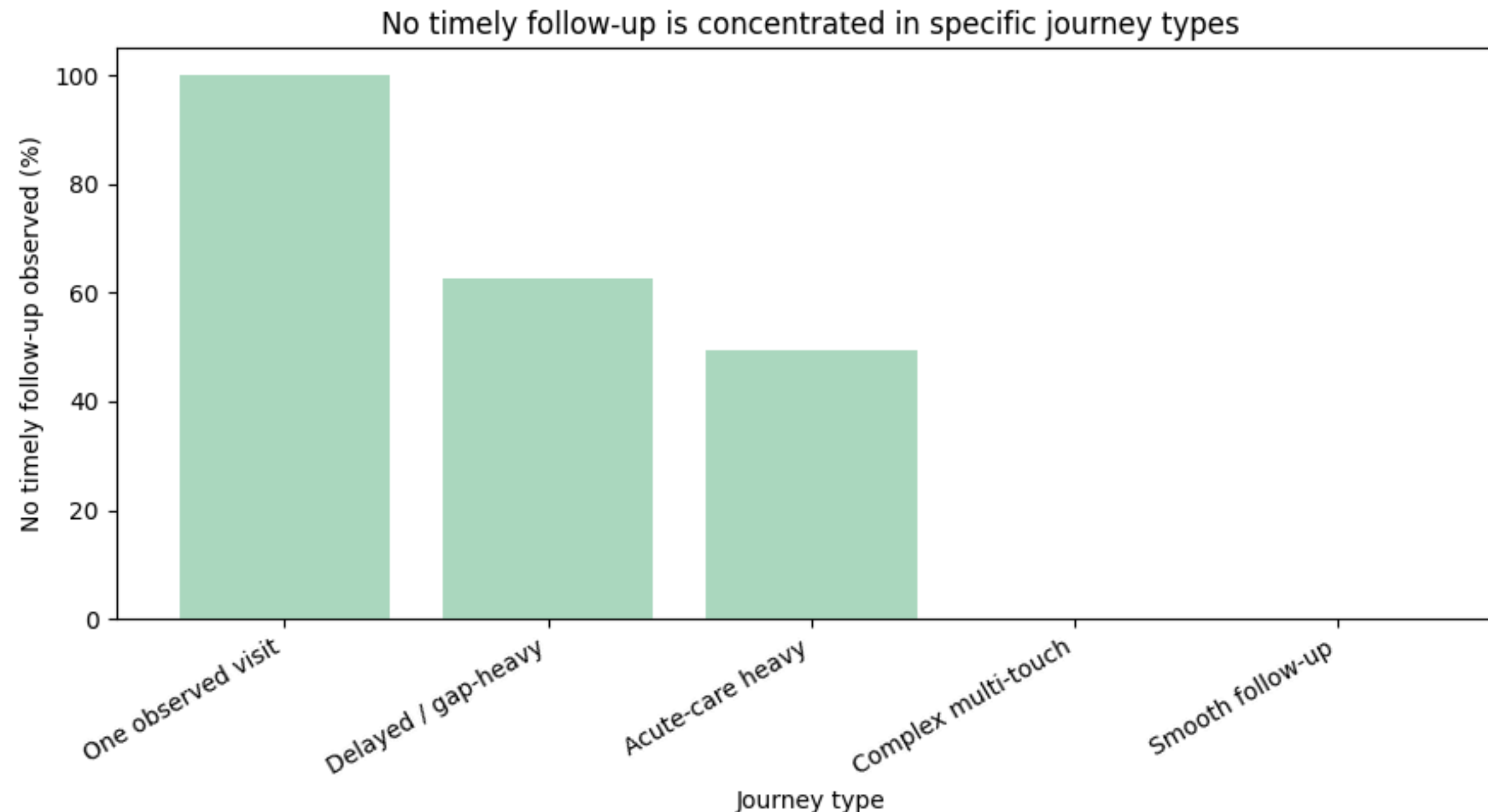
1. Days to first follow-up
2. Journey type
3. Social-risk flags
4. First-visit context

Finding #1: Most diabetes journeys show follow-up gaps



61% no timely follow-up within 4 months

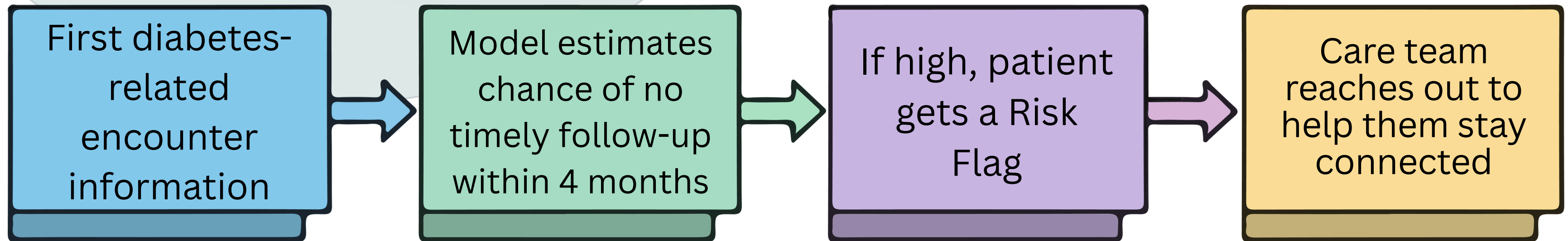
Finding #2: Gap is concentrated, not random



**Care navigation
should target the
first two groups
first.**

Journey Navigation Flag

Predict who may need support by asking “Is this patient at risk of not having a follow-up observed within 4 months?”



Flagged, then what? SVH could do this within 7 days:

1. Call or message the patient for a follow-up appointment.
2. Check transportation barriers.
3. Check whether the patient can use MyChart.
4. Connect them to food, housing, financial, or utility support if needed.
5. Route complex patients to a nurse navigator or care coordinator.

THE PROBLEM

THE FINDINGS

PREDICTIVE MODEL

CONCLUSION

From Reactive care to Proactive care

Before	After
1. Patient has first diabetes encounter	1. Patient has first diabetes encounter
2. System waits	2. System identifies possible follow-up risk
3. Patient may not return for follow-up	3. Care team reaches out early
4. Gap is noticed late, or not noticed at all	4. Patient receives support before the gap becomes larger

- Continuity of care improvement
- Patient experience improvement
- Prioritize who may need outreach first
- Reduces avoidable acute-care use



Thank
You